

Product Overview:

The WIN-SN-TNH-UPOE is a versatile and reliable solution for precise temperature and humidity monitoring, offering robust communication capabilities. It is ideal for measuring environmental conditions in various settings, such as industrial facilities, greenhouses, and smart homes. It provides data transmission via Modbus through an Ethernet interface, ensuring seamless integration into IoT networks. The module offers flexible deployment options tailored to specific monitoring and control needs, ensuring accurate and reliable measurements over long distances with high resistance to electrical interference.



One of the key advantages of the WIN-SN-TNH-UPOE is its flexibility in sensor compatibility, supporting a variety of external sensors via an I2C interface. Powered directly via the network cable, this module eliminates the need for external wall adapters. It boasts **IEEE 802.3at (PoE+) Type 2 Class 4 compliance** with a power capability of **up to 30W**. For maximum deployment flexibility, the device supports a wide input voltage range, seamlessly accommodating both **Active PoE (44V to 57V DC)** and **Passive PoE (38V to 57V DC)** infrastructures.

It features internal galvanic isolation and a robust step-down converter, ensuring safe, reliable performance in diverse industrial IoT applications. The WIN-SN-TNH-UPOE supports the addition of multiple external sensors. This flexibility allows users to customize the module for various applications by integrating sensors suited to different measurement needs. Here are some examples of compatible I2C sensors listed in Table 1.

Technical Specifications:

	Sensor Name	Measuring parameter	Temperature Measuring range	Humidity Measuring range	Measuring Accuracy
☐	AHT21	Temperature, Humidity	-40°C ~ 120°C.	0 – 100 %RH	0.3°C, 2.0 %RH
☒	AHT20	Temperature, Humidity	-40°C ~ 85°C.	0 – 100 %RH	0.3°C, 2.0 %RH
☐	SHT21	Temperature, Humidity	-40°C ~ 125°C.	0 – 100 %RH	0.3°C, 2.0 %RH
☐	SHT25	Temperature, Humidity	-40°C ~ 125°C.	0 – 100 %RH	0.2°C, 1.8 %RH
☐	SHT40	Temperature, Humidity	-40°C ~ 125°C.	0 – 100 %RH	0.2°C, 1.8 %RH
☐	AM2315C	Temperature, Humidity	-40°C ~ 125°C.	0 – 99.9 %RH	0.3°C, 2.0 %RH
☐	BME680	Temperature, Humidity	-40°C ~ 85°C	0 – 100 %RH	±1°C, 3.0%RH
☐	BME280	Temperature, Humidity	-40°C ~ 125°C	0 – 100 %RH	±0.3°C, ±2.0% RH

Table No.1

General Specification:

- Product dimension: 85*85*35 mm
- Active POE Operating Voltage: 44-57V DC
- Passive POE Operating Voltage: 38-57V DC
- Certifications:



Configuration setting:

Communication Speed	Software Configurable
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable via web interface
Function code	0X03 (Read Holding Register)

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40001	0x03	TCP/IP	16 bit int
2	Humidity	40002	0x03	TCP/IP	16 bit int

Additional Feature: SNMP Integration:

In addition to Modbus communication, this module supports SNMP (Simple Network Management Protocol) v2c. This allows an authorized Network Management System (NMS) to directly poll the device for real-time environmental data without relying on Modbus or MQTT push protocols.

SNMP Configuration Settings:

- Protocol Version: SNMP v2c
- Read-Only Community String: Public
- Read-Write Community String: Private
- Enterprise OID Base: .1.3.6.1.4.1.99999

SNMP OID Reference Table:

Because standard SNMP does not natively support floating-point (decimal) numbers, the sensor values are scaled by a factor of 10 before being transmitted. Just like the Modbus registers, you must divide the returned SNMP value by 10 to obtain the true reading.

- **Example:** An SNMP query returning 300 for Temperature equals 30.0°C.

How OIDs Work:

An OID (Object Identifier) functions as a value's specific "address"—much like a house number. When a network manager queries an OID, the sensor node looks up and returns just that specific value.

OIDs form a hierarchical tree, where each number represents a branch. Using the temperature OID (.1.3.6.1.4.1.99999.1.4.0) as an example.

Here is how the address is broken down:

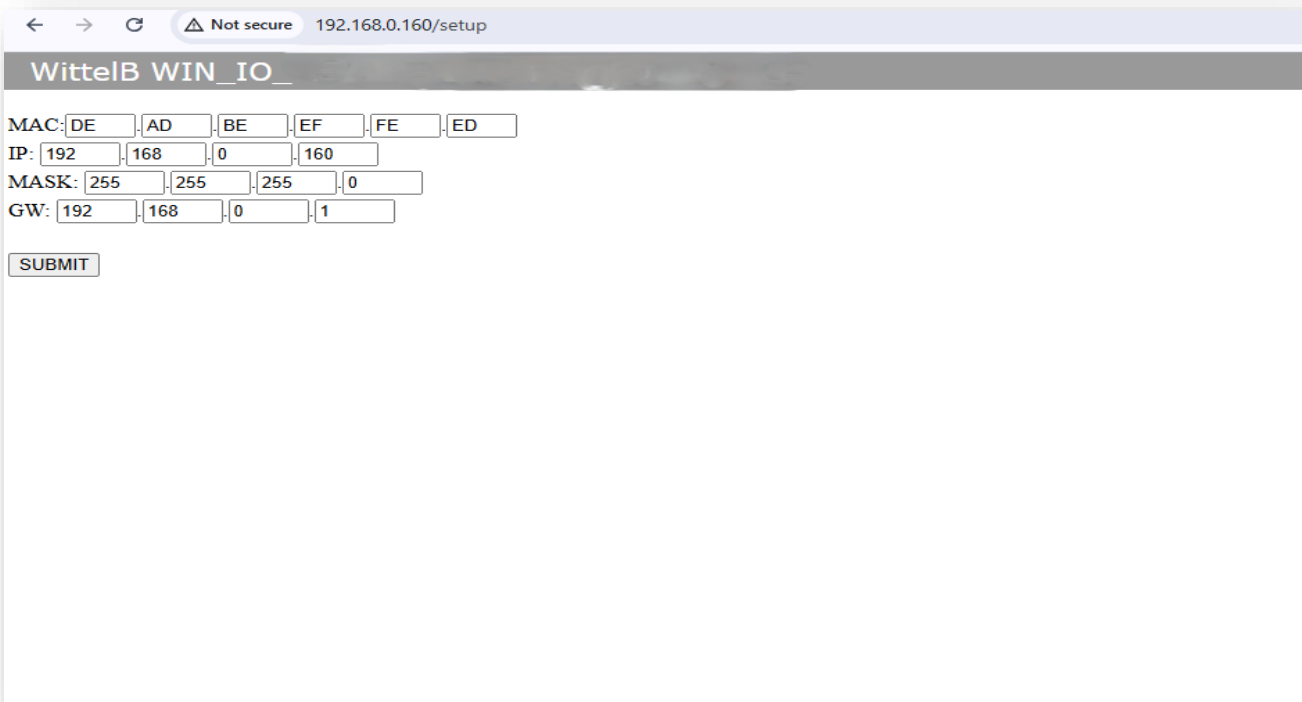
OID Part	Meaning
.1.3.6.1	Standard internet OID prefix — identical on every SNMP device.
.4.1	"Private Enterprise" branch — everything following this is custom, device-specific data.
.99999	Enterprise Number (Placeholder).
.1	Custom sub-branch (e.g., "sensor data").
.4	The specific value in that group (e.g., Temperature).
.0	Marks this as a single scalar value, rather than a list (standard SNMP convention).

WIN-SN-TNH-UPOE OID Mapping:

OID Address	Variable	Meaning
.1.3.6.1.4.1.99999.1.2.0	snmp_ModeFlag	0 = TCP mode, 1 = WiFi mode
.1.3.6.1.4.1.99999.1.3.0	snmp_MqttConnected	1 = Connected, 0 = Disconnected
.1.3.6.1.4.1.99999.1.4.0	snmp_Temperature	Temperature x 10
.1.3.6.1.4.1.99999.1.5.0	snmp_Humidity	Humidity x 10

Configuration Process:-

1. Power up the module by connecting a 5 Vdc 1A power supply and the RED led will be ON. After powering up wait for 20 Seconds.
2. Enter the <http://192.168.0.160/> IP address on a web browser and a webpage with two configuration pages will be accessible.



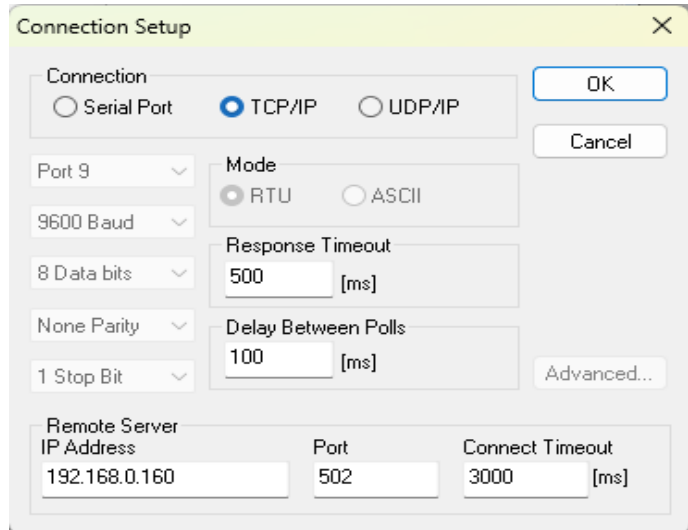
The screenshot shows a web browser window with the address bar displaying "192.168.0.160/setup". The page title is "WittelB WIN_IO_". The form contains the following fields:

- MAC: DE, AD, BE, EF, FE, ED
- IP: 192, 168, 0, 160
- MASK: 255, 255, 255, 0
- GW: 192, 168, 0, 1

A "SUBMIT" button is located below the fields.

3. After submitting it on the webpage, you need to open MB/POLL or MBSCAN on your device. In the connection settings, select TCP/IP Connection and enter the following IP. You will receive data, and for more information, you can refer to the images below.
4. Hardware Fail-Safe: To recover a lost IP address, simply short the onboard RST jumper and press the RESET button to instantly restore factory default configurations.

- **Connection Setup:-**



Connection Setup dialog box showing configuration for TCP/IP connection.

Connection: ☐ Serial Port ☒ TCP/IP ☐ UDP/IP

Port 9 (dropdown)

Mode: ☒ RTU ☐ ASCII

9600 Baud (dropdown)

8 Data bits (dropdown)

None Parity (dropdown)

1 Stop Bit (dropdown)

Response Timeout: 500 [ms]

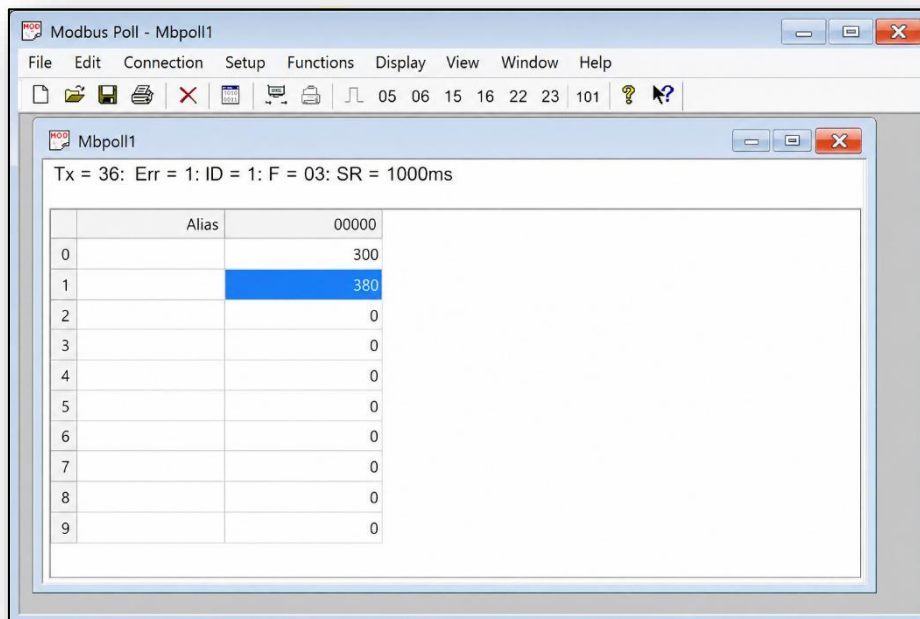
Delay Between Polls: 100 [ms]

Advanced...

Remote Server:

IP Address	Port	Connect Timeout
192.168.0.160	502	3000 [ms]

- **Here is the data**



Modbus Poll - Mbpoll1

Tx = 36: Err = 1: ID = 1: F = 03: SR = 1000ms

	Alias	00000
0		300
1		380
2		0
3		0
4		0
5		0
6		0
7		0
8		0
9		0

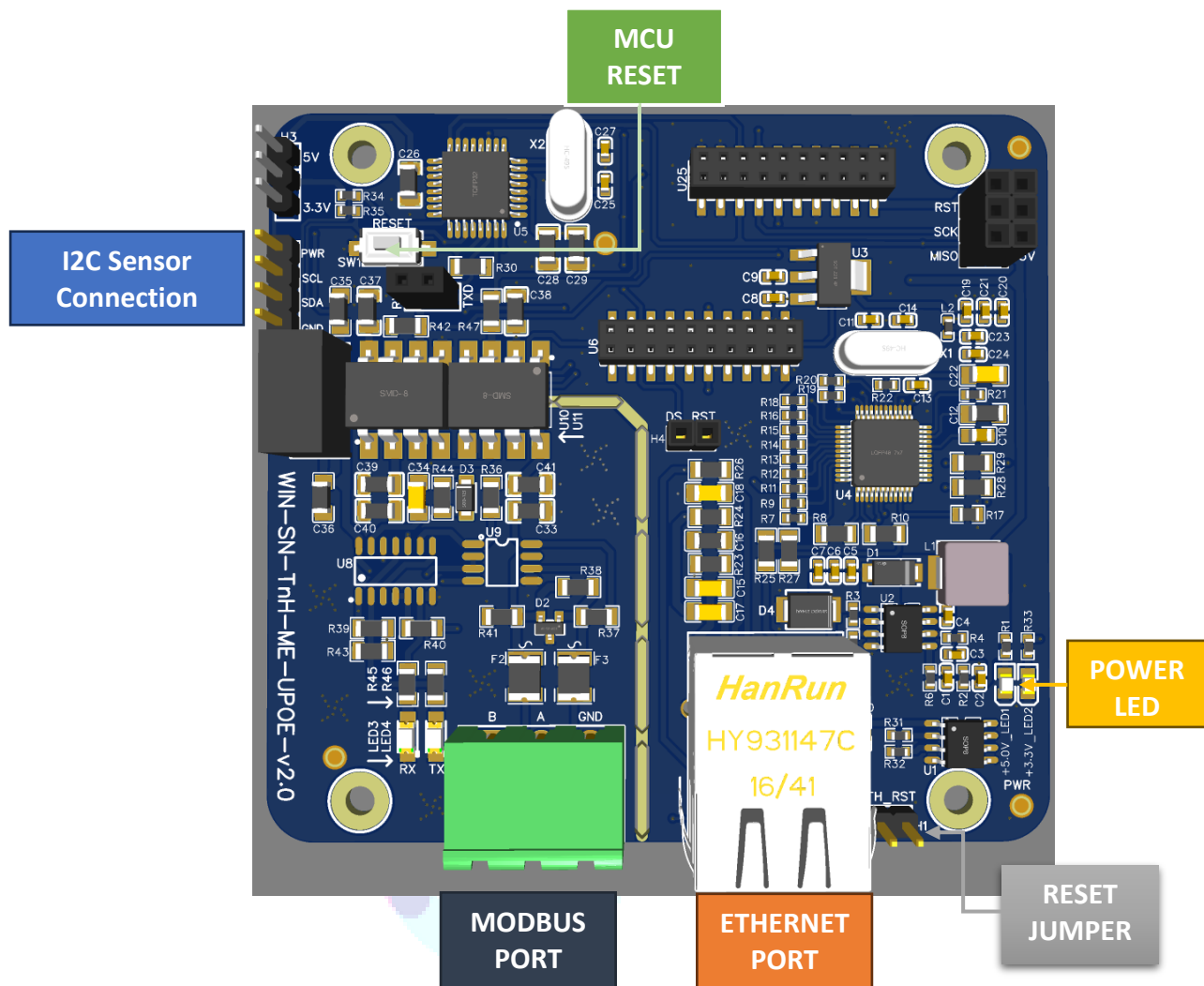
4. Important Note:

Temperature and humidity data are provided in multiples of 10 for higher resolution. To obtain the actual readings, you need to divide the values by 10.

For example:

Temperature: $300 / 10 = 30.0^{\circ}\text{C}$

Humidity: $380 / 10 = 38.0\%$



Contact Information

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